Manan Chahal

AI/ML & Software Development | Final-year Computer Engineering

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Profile

Final-year Computer Engineering student interested in building scalable AI/ML products, with hands-on experience developing computer vision, retrieval-based and geospatial applications. Industry experience at NSE working on AI-based translation, RAG and model deployment experiments, alongside end-to-end AI projects spanning machine learning, backend development and deployment.

Experience

AI/ML Intern – National Stock Exchange of India (NSE) May 2026 – July 2026

- Worked on an automated multilingual translation system using Sarvam AI, server-side rendering, DOM traversal and translation batching.
- Explored Redis and Elasticsearch caching using sentence hashing and cache invalidation for translated-content retrieval.
- Contributed to an internal ChatGPT-style assistant using Next.js, Redis and RAG, deployed on Linux.
- Implemented speaker diarisation using pyannote and explored ONNX/SafeTensors conversion for model deployment.
- Explored the feasibility of using QLoRA to fine-tune Voxtral for real-time ASR for Marathi through a small-scale experiment.

Selected Projects

AI-Powered FSSAI Packaging Compliance Platform *Python, Computer Vision, OCR, RAG, FastAPI*

- Built an AI-based system that analyses product-packaging images to extract relevant information and identify potential FSSAI packaging violations.
- Developed a computer-vision and OCR pipeline to identify key regulatory packaging elements including claims, ingredients, nutrition information and statutory details.
- Integrated regulatory retrieval and rule-based checks to validate extracted information against relevant FSSAI requirements.
- Placed third in a college hackathon; the project continues to be developed further.

AI-Powered Civic Issue Reporting Platform *Python, PyTorch, OpenCV, Node.js, Express.js, Leaflet.js*

- Fine-tuned a ResNet-34 model to verify complaint images before submission, aimed at reducing false reports.
- Built an interactive geospatial interface using Leaflet.js with distance-based dynamic loading of complaints.
- Developed backend APIs using Node.js and Express.js, including rate limiting to mitigate spam.

Glacial Flow Lines Detector (CSRE, IIT Bombay) *Python, Random Forest, QGIS, PolSARpro*

- Built a Random Forest classifier for the MW Satellite Data Processing course to identify glacier flow lines from microwave satellite imagery.
- Processed SAR datasets using QGIS and PolSARpro to generate interpretable geospatial outputs.

Cloud-Based E-Learning Platform *React, Express.js, PostgreSQL, Azure, Linux, Nginx*

- Built a full-stack web application using React and Express.js with REST APIs and authentication middleware.
- Deployed the application on Microsoft Azure Linux VMs with SSL termination through an Nginx reverse proxy.

Education

2023 – 2027	B.Tech in Computer Engineering, **Vidyalankar Institute of Technology** University of Mumbai	CGPI: **9.87**
2023	Class XII (CBSE) – Green Fingers Global School	85.8%
2021	Class X (ICSE) – Ryan International School	85.4%

Achievements

- Secured **First Place** in Soft Computing, Semester VI examinations.
- **1st Place** CodeSprint (CSI-VIT) | **2nd Place** Bug Auction.
- **3rd Place** Jeopardy (ITSA-VIT) | **3rd Place** Cognitive Clash College Level Hackathon.
- Solved **350+** problems on LeetCode and competitive-programming platforms.

Technical Skills

Languages	Python, Java, JavaScript, C
AI / ML	PyTorch, scikit-learn, OpenCV, LLMs, RAG, Speaker Diarisation, ONNX
Web / Data	React, Next.js, Node.js, Express.js, Redis, PostgreSQL, MongoDB, Elasticsearch
Cloud / DevOps	Azure, Linux, Git, GitHub, Nginx, CI/CD